Bhagyashree Mohalkar

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PORTFOLIO | LINKEDIN | GITHUB | HACKERRANK | CODECHEF

EDUCATION

Bachelor of Technology in Electronics and Computer Engineering CGPA: 8.74 **Aug 2023 – Present**
Honors in Artificial Intelligence and Machine Learning
Sanjivani College of Engineering, Kopargaon, Maharashtra

Higher Secondary Certificate (HSC) **Jan 2022 – Feb 2023**
R.B. Narayanrao Borawake College, Shrirampur

Secondary School Certificate (SSC) **Jun 2020 – Jul 2021**
De Paul English Medium High School, Shrirampur

INTERNSHIPS

Project Work Intern | ***Bhabha Atomic Research Centre (BARC), Mumbai (On-site)*** **Apr 2026 – July 2026**
- Re-architected pyDOSEIA (radiological dose assessment software) into a FastAPI, PostgreSQL, and JWT-based **client-server** platform supporting **100+** concurrent users.
- Curating and preprocessing a **130k+** molecule quantum chemistry dataset with DDEC4 atomic charge labels to train an **E(3)-equivariant Graph Neural Network** for atomic partial charge prediction.

Artificial Intelligence Intern | ***Infosys Springboard Virtual Internship 6.0*** **Nov 2025 – Jan 2026**
- Completed hands-on AI modules covering supervised learning, model evaluation, and end-to-end deployment workflows.

GenAI Powered Data Analytics Job Simulation | ***Tata Group (Virtual, via Forage)*** **Jul 2025**
- Completed practical tasks in exploratory data analysis, risk profiling, and predicting delinquency with AI, and built a data-driven collections strategy with AI-driven implementation and business storytelling.

Data Analytics Job Simulation | ***Deloitte (Virtual, via Forage)*** **Jun 2025**
- Completed practical tasks in data analysis and forensic technology as part of Deloitte's Data Analytics job simulation program.

PROJECTS

LnQM Charge Prediction | *Equivariant GNN for Molecular Partial Charge Prediction* | *BARC* **Jul 2026**
- Prepared the **DDEC4 molecular dataset** for training an **Equivariant Graph Neural Network (GNN)**, handling dataset preprocessing, graph data preparation, and feature engineering.
- Adapted the training pipeline for a Linux-based computational environment, gaining hands-on exposure to machine learning workflows for molecular partial charge prediction.
- **Technologies:** Python, PyTorch, PyTorch Geometric, e3nn, NumPy, HDF5/h5py, RDKit, Linux, TensorBoard

pyDOSEIA | *Client-Server Radiological Dose Assessment Platform* | BARC **April 2026**
- Re-architected a radiological dose assessment application into a scalable **client-server platform**, developing backend services and **10+ REST APIs** with FastAPI and PostgreSQL to support secure multi-user access for **2000+ concurrent users**.
- Implemented **JWT-based authentication, RBAC, and bcrypt password hashing**, along with persistent file management, **500 MB/user storage quotas**, and **365-day retention**, and integrated the platform with a Streamlit frontend on **Ubuntu Linux**.
- **Technologies:** Python, FastAPI, PostgreSQL, REST APIs, JWT, bcrypt, Streamlit, Ubuntu Linux

TraceFinder | *Forensic Scanner Identification System* | Self Project **Dec 2025**
- Developed a **CNN-based image classification system** for forensic scanner identification, training on **3,000+ scanned documents** and achieving **94% classification accuracy**.
- Engineered an image preprocessing pipeline using **Python and OpenCV** to extract scanner-specific visual and noise patterns, and evaluated CNN and traditional ML classifiers using **accuracy, precision, recall, F1-score, and confusion matrix**.
- **Technologies:** Python, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, Convolutional Neural Networks (CNN), Image Classification, Machine Learning, Computer Vision

SKILLS

Programming Languages: Python, C, JavaScript

Web Development: HTML, CSS, FastAPI, Flask, REST APIs, JWT Authentication, Object-Oriented Programming (OOP), MySQL, PostgreSQL

AI & Machine Learning: NumPy, Pandas, PyTorch, TensorFlow, Scikit-learn, Convolutional Neural Networks (CNN), Graph Neural Networks (GNN), Natural Language Processing (NLP)

Data & Visualization: Statistical Analysis, Data Visualization, Power BI, Streamlit

Tools & Platforms: Git, Linux/Unix CLI

Soft Skills: Teamwork, Adaptability, Attention to Detail, Problem Solving

EXTRACURRICULAR ACTIVITIES

President | *Student Alumni Relation Cell (SARC), Sanjivani College of Engineering* **Aug 2025 – May 2026**

- Led alumni interaction events and networking drives to strengthen student engagement and support placement initiatives.

Training & Placement Student Coordinator | *Sanjivani College of Engineering* **Aug 2024 – May 2026**

- Coordinated campus placement activities and built connections between industry professionals and students.

Team Lead, Space & Robotics | *Team Charlie, Sanjivani College of Engineering* **Sep 2023 – Dec 2024**

- Promoted from Executive Member to Team Lead of the Space and Robotics vertical; built projects and conducted workshops in space technology and Automation.

RELEVANT COURSEWORK

- Foundation of Data Science
- Advance Data Science
- Supervised Modelling and AI Technologies
- Deep Learning
- Discrete Mathematics and Information Theory
- Mini Project: Artificial Intelligence and Machine Learning
- Database Management System
- Cloud Computing
- Computer Networks and Security
- System Programming and Operating System
- Electronic Devices and Circuits
- Microprocessor & Microcontroller

CERTIFICATIONS

- Completed the **Data Science Methods & Algorithms** certification from **Udemy** (2026). LINK
- Completed the **Honeywell Future Edge Skill Program on Artificial Intelligence** (2026). LINK
- Completed the **Internet of Things (IoT)** certification offered by **DADB India** (2026). LINK
- Earned the **Google Cloud Career Launchpad Cloud Engineer track** (2025). LINK
- Completed the **NPTEL – Project Management** certification (2025). LINK
- Completed the **Data Science with Python (Capsule Course)** certification (2023). LINK

COMPETITIONS & ACHIEVEMENTS

- Registered a design patent titled "**Hybrid Solar-Wind Energy Harvesting Pole System**" under the **Designs Act, 2000** (2026). LINK
- Registered a design patent titled "**IoT-based Smart Energy Meter with Alert System**" under the **Designs Act, 2000** (2026). LINK
- Registered a design patent titled "**Blind Spot Detection System for Vehicles**" under the **Designs Act, 2000** (2026). LINK
- Ranked in the **Top 5%** nationwide and awarded the **Elite + Silver** certification in **NPTEL – Enhancing Soft Skills and Personality** (Score: **84%**, 2025). LINK
- Qualified for the **Case Study Round** of **WWT Unravel 2025**.
- Registered a design patent titled "**Precision Control Brake with Ergonomic Grip**" under the **Designs Act, 2000** (2024). LINK
- Secured **2nd Place** in the **Department of Electronics & Computer Engineering Technical Project Competition & Exhibition** for the project "**Columbus Voice Automation System**" (2023). LINK
- Secured **124th Rank** nationally in the **National Level Genius in Science and Maths Olympiad** (2017). LINK
- Participated in the **National Science Olympiad** (2015). LINK